Weirui Wang

wangwr2024@shanghaitech.edu.cn · (+86) 15662661930

Education

ShanghaiTech University, Shanghai, China — Sep. 2024 – Jun. 2027 (Expected)

M.S. in Computer Science
Advisors: Prof. Jingyi Yu and Prof. Lan Xu

Shandong University, Jinan, China — Sep. 2020 – Jun. 2024

B.Eng. in Control Science and Engineering
Overall Grade: 91.57/100

Publications

- ACT: A Unified Framework for Rigging and Animating Characters with Arbitrary Topologies

  ACM TOG (SIGGRAPH 2026)

  Pengyu Long, **Weirui Wang****, *Qingcheng Zhao, Xiaoyang Guo, Xiaoyu Pan, Qixuan Zhang, Jiaqing Zhou, Tianlei Hu, Wei Yang, Lan Xu, Jingyi Yu* (* Equal contribution)

Research Experience

ShanghaiTech University, Shanghai, China — Sep. 2024 – Present

Master's Student Researcher, advised by Prof. Jingyi Yu and Prof. Lan Xu

ACT: A Unified Framework for Rigging and Animating Characters with Arbitrary Topologies

ACM TOG / SIGGRAPH 2026, Co-first Author

Observed a gap between professional animation practice and existing automatic pipelines. Artists repeatedly test motions and refine joint placement and skinning until a character deforms correctly, whereas most methods infer a rig from static geometry and generate motion only afterward. As a result, a skeleton may fit the character at rest but lack the articulation needed for its intended actions, causing deformation errors during animation.

Originated ACT's unified formulation after initially developing the motion branch, and translated it into the core model and training pipeline. ACT represents the rest-pose skeleton and temporal joint trajectories in a shared spatiotemporal space, allowing motion to guide joint placement and skinning from the beginning, while the generated rig constrains the motion. The framework jointly produces skeletons, skinning weights, and text-driven motions for arbitrary character topologies, while preserving explicit and editable assets for standard animation workflows.

AniRig: Recovering Animation-Ready Rigged Assets from Arbitrary-Topology 3D Characters

Submitted to SIGGRAPH Asia 2026, First Author

Generated or scanned 3D characters often arrive already posed and contain only a surface mesh. To animate them, artists still need a neutral rest-pose mesh, skeleton, and skinning weights, which usually require substantial manual work.

Proposed AniRig to recover all three from a single posed mesh. By jointly reasoning about the visible pose and the underlying rest structure, the method handles folded or touching body parts and produces editable assets for re-posing and animation reuse. Led the project end to end.

Industry Experience

Deemos Technologies Inc., Shanghai, China — Apr. 2024 – Present

Research Intern

- **Facial Texture Enhancement:** Developed an identity-preserving enhancement method for generated 3D characters. Combined face localization, reference-guided inpainting, geometric constraints, and super-resolution to improve facial detail while preserving identity and the original 3D structure.
- **Automatic Character Rigging:** Adapted ACT to real-world 3D assets with complex topology, broken surfaces, and irregular geometry. Developed validation and failure-handling methods to improve the robustness of skeleton and skinning prediction. The resulting methods are used in production at Hyper3D and ByteDance.